

Batch: A -4 **Roll No.:** 16010122151

**Experiment / assignment / tutorial
No. 1**

**Grade: AA / AB / BB / BC / CC / CD
/DD**

**Signature of the Staff In-charge
with date**

TITLE: To understand the trouble of interacting with products/machines in day-to-day activity.

Objective:

Identify the problems faced by a user while using products in day-to-day work. This will help in improving the interface in enhancing user experience.

Expected OUTCOME of Experiment: CO 2

To design a more seamless and less frustrating interaction experience for everyone with the technologies they encounter daily.

Books/ Journals/ Websites referred:

< The students should write clearly the books, websites or journal papers, and articles they have refereed to while completing the task >

Background Theory:

Usability issues with everyday products and machines often stem from poor design choices that neglect user needs, leading to difficulties like confusing interfaces, lack of feedback, inconsistent controls, or accessibility barriers. From an HCI and UX perspective, these problems arise due to insufficient understanding of user behavior, cognitive load, and the context in which products are used. To address these challenges, designers should adopt user-centered approaches such as iterative prototyping, usability testing, and feedback loops to create intuitive, responsive, and inclusive experiences. Building observational skills—like noticing user errors, hesitations, or workarounds in real-world settings—can reveal critical design flaws and opportunities for improvement. Ultimately, prioritizing empathy and functionality in design empowers engineers to develop products that truly serve and delight their users.

Effective interaction with products and machines is fundamental to their utility and user satisfaction. Computer engineering students often focus on the technical implementation of systems, but understanding how users interact with these systems is equally crucial. This lab explores the concept of "usability," which refers to the ease of use and learnability of a human-made object. Poor usability can lead to frustration, errors, inefficiency, and even safety hazards.

Key concepts to consider include:

- **Affordances:** The perceived and actual properties of a thing, primarily those fundamental properties that determine just how the thing could possibly be used. (e.g., a handle affords grasping).
- **Signifiers:** Any mark or sound, any perceivable indicator that communicates appropriate behavior to a person. (e.g., a "PUSH" sign on a door).
- **Mapping:** The relationship between controls and their effects. Good mapping leads to intuitive operation (e.g., steering wheel turning a car).
- **Feedback:** Sending information back to the user about what action has been done, what has been accomplished, and what is happening. (e.g., a click sound when a button is pressed).
- **Conceptual Model:** The user's mental model of how a system works. A good design helps users form an accurate conceptual model.
- **Visibility:** The state of the system and its possible actions should be clearly visible to the user.
- **Consistency:** Similar operations should have similar sequences of actions and similar elements should look similar.
- **Error Prevention and Recovery:** Designs should prevent errors where possible and provide clear ways to recover when errors occur.
- **User-Centered Design (UCD):** An iterative design process in which designers focus on the users and their needs in each phase of the design process.

Many products, despite their advanced functionality, suffer from poor design that leads to common interaction troubles. These can range from confusing interfaces on a remote control to poorly labeled buttons on a washing machine, or complex setup procedures for smart devices. This lab encourages you to observe and document these real-world examples, fostering a critical eye for design.

Laboratory Work:

Activity:

1. Choose any two different everyday products or machines that you interact with regularly or can observe easily.

Examples: a coffee machine, a TV remote, a microwave oven, a public vending machine, , a smart thermostat, or even a door handle.

2. While using these machine imagine or observe and note separately:

- Where do you (or others) hesitate?
- Where do you make mistakes?
- What causes confusion?
- What takes longer than it should?
- Are there hidden functionalities?
- Is the feedback clear?
- Are controls intuitive?
- Does the product behave as expected?
- Are there any wrong cues?

3. Document each interaction tabulate meticulously:

Name of the Product under consideration	TV Remote
Reasons for hesitation	Too many buttons, user pauses to search for the right one.
Reasons for mistakes	Buttons look similar
Source for confusion	Icons/symbols (e.g., AV/HDMI) not clear to non-technical users.
Time required to complete the task compared while using the system / interface	☐ Less ☐ Same (no change) ☒ More
Help and assistant	☐ Not easily available → user learns by trial and error.
Product worked as expected	☐ Sometimes yes, but wrong presses often create unexpected results.

4. Identify common themes or recurring types of usability issues across different products.
 - **Overloaded interface** → too many small buttons.
 - **Unclear symbols** → difficult for elderly or first-time users.
 - **Weak feedback** → just a blink light, no confirmation if input worked.
5. Reflect on how these issues could have been prevented during the design and engineering phases.

Too Many Buttons (Overloaded Interface)

Prevention: Designers could have followed the principle of **simplicity & minimalism**, keeping only frequently used buttons (power, volume, channel, input, menu). Less-used features (record, apps, special modes) could be moved to an on-screen menu instead of the physical remote.

Unclear Symbols and Icons

Prevention: Standardized **labels and signifiers** (e.g., “HDMI” spelled out instead of just a cryptic icon) would reduce confusion. Using **universally recognized icons** and color coding could also make controls more intuitive.

Weak Feedback

Prevention: Remote could provide **clearer feedback** (like an on-screen confirmation pop-up: “Volume Changed”, “Source Selected”). Haptic feedback (tiny vibration) or distinct sound feedback could also confirm correct button presses.

Cluttered Button Layout / Wrong Cues

Prevention: Better **mapping and grouping** of buttons → e.g., volume & channel controls separated and highlighted with distinct shapes or colors. Following **consistency across brands** would help users transfer knowledge from one remote to another.

Accessibility Issues

Prevention: Designers could have applied **user-centered design (UCD)** by testing prototypes with elderly users, children, and non-technical users to identify usability problems early. Larger buttons, high-contrast labels, and backlighting could make remotes easier for everyone.

6. Document each interaction tabulate meticulously:

Name of the Product under consideration	Microwave Oven
Reasons for hesitation	User hesitates when selecting power levels or modes (Grill/Defrost/Auto-cook).
Reasons for mistakes	Wrong time entered due to confusing keypad/rotary knob; some users accidentally press "Start" without setting time.
Source for confusion	Icons on buttons (snowflake = defrost, waves = heat) not universally understood.
Time required to complete the task compared while using the system / interface	○ Less ○ Same (no change) ● More
Help and assistant	☐ Not intuitive; users often need to refer to manual or guess.
Product worked as expected	Sometimes yes; advanced modes often confuse users.

7. Identify common themes or recurring types of usability issues across different products.

- **Complex controls for basic tasks.**
- **Unclear icons/symbols** that don't communicate purpose.
- **Inconsistent behavior** between brands → users carry wrong assumptions.
- **Weak feedback** → a beep doesn't always confirm *what* happened.

8. Reflect on how these issues could have been prevented during the design and engineering phases.

Most issues could be prevented during design and engineering by applying user-centered design principles:

- Simplify controls → Provide a clear “Quick Heat” option for everyday use, while grouping advanced modes under a separate menu.
- Use clear labels → Replace confusing icons with text labels (e.g., “Defrost” instead of a snowflake).
- Improve feedback → Display messages like “Heating Started” or “Defrost Mode Activated” on screen instead of just beeping.
- Consistency → Standardize button functions across models so users don’t relearn every new microwave.
- Accessibility → Larger, well-lit buttons and voice guidance for elderly users.

Post Laboratory Questions:

Following questions are to be handwritten and attach photograph of the answer in this document.

1. What were the top three most common types of interaction troubles encountered?

- Overloaded interface: Too many buttons and options leading to hesitation and errors.
- Unclear symbols and labels: Icons (like disc, snowflake, HDMI) often misinterpreted or unknown to users.
- Weak or ambiguous feedback: Simple beeps or lights without clear confirmation of the action taken.

2. Impact of poor design in user experience.

Poor design can lead to frequent user mistakes, frustration, increased task time, reduced efficiency, and ultimately, avoidance of the product or negative brand perception.

3. Define User Experience (UX) in your own words, distinguishing it from User Interface (UI).

User Experience (UX) is the overall feeling and effectiveness of using a product—how easily and pleasantly tasks can be completed. In contrast, User Interface (UI) refers to the visual and interactive elements (buttons, layout, icons) users interact with. UX is about the overall journey and satisfaction; UI is one part of how that journey is presented.

4. Name and briefly describe two key components or aspects that contribute to a positive user experience.

- Usability: The product should be easy to understand and use, with intuitive navigation and controls.
- Feedback & responsiveness: Immediate and meaningful responses to user actions—visual, auditory, or haptic—confirm that things are working as expected.

5. Why is understanding the user's context and emotions important when designing for user experience?

Understanding context and emotions ensures the design meets real user needs and scenarios—whether hurried, stressed, elderly, or novice. It helps create empathetic, adaptable, and supportive experiences that align with how users actually behave and feel.

6. How do usability and accessibility contribute to the overall user experience of a product?

- Usability ensures tasks are completed efficiently and intuitively, reducing errors and cognitive load.*
- Accessibility ensures that users of all abilities—including those with visual, motor, or cognitive impairments—can use the product effectively, promoting inclusivity and better overall satisfaction.*